SC4001 · Part B

YourName | Neural Networks and Deep Learning | October 3, 2026

Generative AI declaration

I used OpenAI Codex / ChatGPT for planning, code generation/debugging, experiment execution, analysis, figures, report drafting and submission packaging. Verification included assignment/library-source checks, 96 behavioral tests, CUDA checkpoint restoration, saved-prediction metric/BCE checks, independent parameter/FLOPs formulas and artifact hashes. Grad-CAM tensors were reproduced by independent PyTorch autograd. All results below come from completed saved experiments. I remain responsible for reviewing and understanding the submitted work.

Data, architecture interpretation and common protocol

Use the supplied 4,708/524/624 train/validation/test partitions and 64×64 grayscale images. Divide uint8 by 255 once, then normalize with training-pixel mean 0.5716994213 and population SD 0.1740975398. Numeric labels are assumed 0=healthy, 1=pneumonia; the archive has no semantic metadata. Eight distinct images overlap train/validation; three extra copies occur within test. No exact train/test or validation/test matches were found, but patient independence cannot be checked without IDs.

Convolutions use SAME padding, no bias, then BN/ReLU; FC layers have bias; residual sums use ReLU. Interpret the caption's 'all layers' as hidden layers: final FC1 has an unrestricted logit with BCEWithLogitsLoss; sigmoid≥0.5 predicts class 1. Adam, seed 42, 20-epoch cap, patience 4 and loss improvement 0.0001 apply throughout. Stop on validation loss; save lowest validation classification error, then lower loss, then earlier epoch. Across tied candidates, prefer the smaller rate/batch. No augmentation, weighting, resampling or refit; test is evaluated once per selected model. Metrics use full-partition evaluation mode.

B1. AlexNet-like learning-rate selection

Figure B1 is implemented as 5×5/16 → max pool → 5×5/32 → max pool → 3×3/64 → max pool → average output 4×4 → FC64 → FC32 → FC1, with dropout 0.5 only in hidden FC layers. All five rates use batch 128 and fresh initialization; 99,569 trainable parameters.

Learning rate	Best epoch	Val. BCE loss	Val. error	Sensitivity	Specificity
1e-05	18	0.283130	9.5420%	97.1722%	71.1111%
1e-04	19	0.076439	3.0534%	97.4293%	95.5556%
1e-03	**9**	**0.064911**	**2.0992%**	**98.2005%**	**97.0370%**
1e-02	5	0.199377	5.9160%	98.7147%	80.7407%
1e-01	17	0.103817	4.5802%	96.1440%	93.3333%

Select LR=0.001, epoch 9: 11/524 validation errors (97.9008% accuracy), versus 16 for the next-best rate 0.0001. Every metric in a row belongs to that one saved checkpoint; stopping loss and error selection can prefer different epochs. The frozen learning rate supplies B2 and B3.

Why sensitivity and specificity are necessary

Validation is 74.24% positive: predicting pneumonia for everyone would yield 74.24% accuracy, 100% sensitivity and 0% specificity. Sensitivity=TP/(TP+FN) measures detected positives; specificity=TN/(TN+FP) measures correctly rejected negatives. LR 0.01 has higher sensitivity than 0.001 but far lower specificity (80.74% versus 97.04%), a trade-off hidden by one aggregate score.

B1 · learning curves and test; B2 selection table

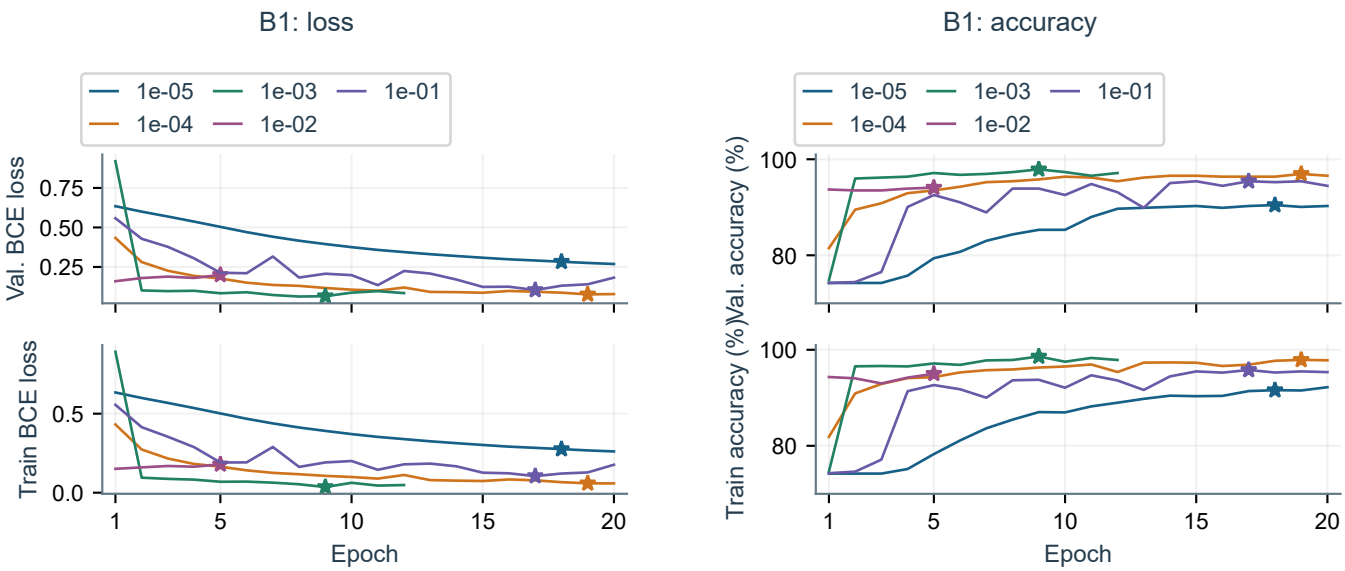


Figure B1a–b. Separate loss and accuracy figures, each with all five validation and training curves. Stars mark saved checkpoints. Accuracy axes start at 70%; curves end at actual stopping epochs.

LR 1e-5 learns slowly: loss keeps falling at the 20-epoch cap, but best accuracy is only 90.46%. 1e-4 improves more gradually to 96.95% at epoch 19. 1e-3 reaches minimum loss at epoch 8, but epoch 9 has lower error; patience stops at 12. Its selected train/validation losses are 0.035620/0.064911. 1e-2 worsens in both train/validation loss and stops at 5, indicating an unfavorable optimization trajectory rather than a simple overfitting pattern. 1e-1 shows reversals and reaches only 95.42% best accuracy. Finite-budget stopping is not proof of global convergence; BN/dropout causes were not isolated.

B1. One final test evaluation

Test accuracy	Sensitivity	Specificity	TP / TN / FP / FN
85.4167%	99.7436%	61.5385%	389 / 144 / 90 / 1

The epoch-9 checkpoint misses one positive but labels 90/234 negatives positive. Test accuracy is 12.48 percentage points below validation, chiefly with specificity dropping to 61.54%. Prevalence changes affect accuracy but cannot alone explain lower specificity. Supplied overlap/dataset differences may contribute; their effects are not identified. No settings or threshold were changed after test.

B2. VGG-like batch-size selection, frozen LR=0.001

Batch size	Best epoch	Val. BCE loss	Val. error	Sensitivity	Specificity
16	**13**	**0.069390**	**2.0992%**	**98.9717%**	**94.8148%**
32	13	0.090230	2.8626%	96.9152%	97.7778%
64	6	0.097501	2.8626%	98.7147%	92.5926%
128	5	0.135308	4.7710%	98.7147%	85.1852%
256	16	0.085114	2.8626%	98.2005%	94.0741%

Batch 16 wins uniquely with 11 errors, versus 15 for 32/64/256 and 25 for 128. Batch 32 has higher specificity, but the prescribed lowest-error criterion selects batch 16. LR/batch are frozen for B3.

B2 · batch-size curves, interpretation and final test

The VGG-like model has two 3×3 Conv/BN/ReLU layers per width 8,16,32, each pair followed by max pooling, then global average pooling and FC1: 18,185 parameters, no dropout. All candidates share initial tensors and the underlying shuffled row sequence; batch grouping changes.

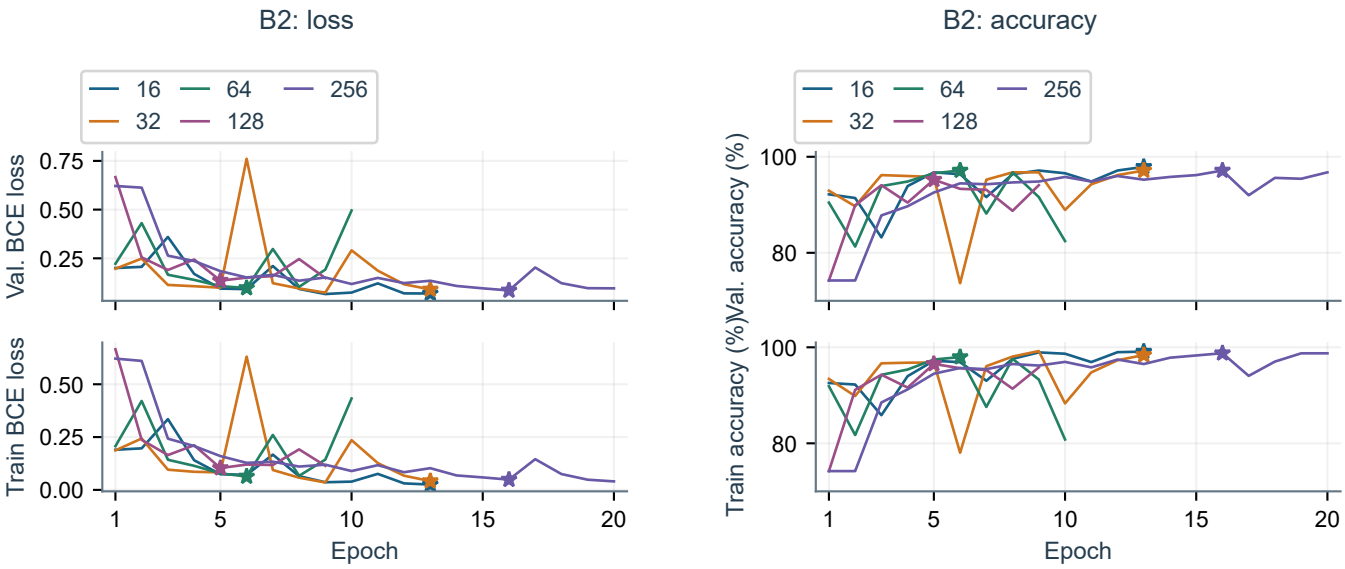


Figure B2a–b. Separate loss and accuracy figures for all five batches; validation above training. Stars mark error-selected checkpoints, not necessarily the minimum-loss epoch.

Batch 16 reaches minimum validation loss at epoch 9 (0.066596), but lowest error at 13 (0.069390); loss patience expires at 13. Its train/validation accuracies are 99.11%/97.90%. Batch 32 has a large epoch-6 loss spike/accuracy drop and then recovers, selecting 13. Batch 64 selects 6, then deteriorates through stopping at 10. Batch 128 begins near the positive-class majority rate, selects 5 and stops at 9. Batch 256 starts slowly, selects 16, and exhausts patience at the 20-epoch cap. Simultaneous train/validation reversals do not justify attributing every spike to overfitting.

Batch size also changes the update budget

For batches 16/32/64/128/256, updates per epoch are 295/148/74/37/19. Selected checkpoints receive 3,835/1,924/444/185/304 updates; batch 16 receives 12.6× as many as batch 256. Smaller batches alter gradient/BN estimates, while larger batches make fewer updates per epoch. Gradient variance and BN effects were not separately measured. This one-seed comparison selects the required training recipe, not a universal batch-size ranking or an equal-update experiment.

B2. One final test evaluation

Test accuracy	Sensitivity	Specificity	TP / TN / FP / FN
85.5769%	99.2308%	62.8205%	387 / 147 / 87 / 3

The epoch-13 checkpoint misses three positives but makes 87 false positives. Specificity drops from 94.81% validation to 62.82% test; accuracy falls 12.32 percentage points. Low specificity is the main observed weakness. Conditional specificity is not explained by prevalence alone. The semantic-label assumption, supplied train/validation overlap and absent patient IDs limit interpretation; partitions and frozen choices were preserved.

B3 · compact ResNet and architecture comparison

Use LR=0.001 and batch 16 exactly from B1/B2. Figure B3 has an initial 3×3/16 stage/max pool, then residual blocks at 16/32/64 channels; downsampling blocks use stride-2 main convolutions and 1×1 projections, with ReLU after sums. Global averaging feeds FC1. Seed 42 selects epoch 9: validation accuracy 97.5191%, loss 0.061988; loss stopping ends at 13.

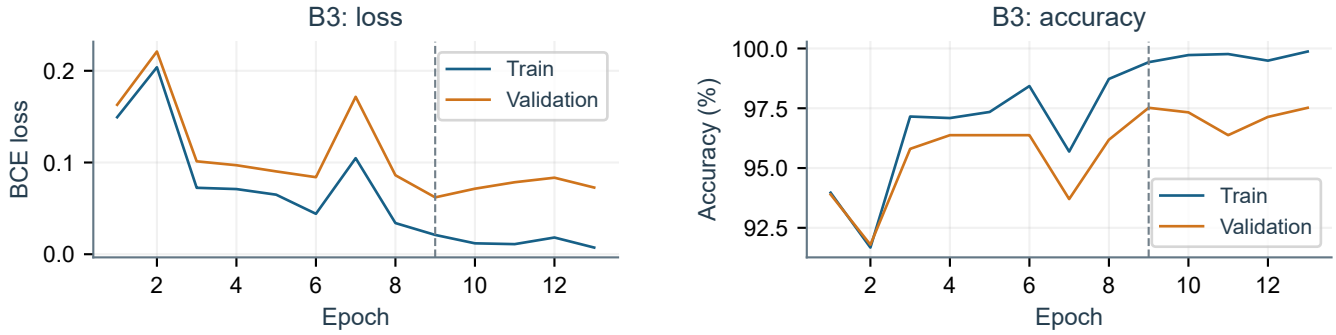


Figure B3a–b. Training/validation curves; dashed line marks selected epoch 9. The growing later gap is consistent with overfitting; earlier simultaneous fluctuations have unisolated causes.

At epoch 9, train/validation accuracies are 99.43%/97.52%; by epoch 13 training reaches 99.87% while validation does not improve. The final test detects 389/390 positives but misclassifies 106/234 negatives. Specificity is 54.70%. The table reuses B1/B2 saved final tests without re-evaluation.

Model	Accuracy	Sensitivity	Specificity	Class. error	Parameters	FLOPs/image
B1 AlexNet	85.4167%	99.7436%	61.5385%	14.5833%	99,569	39,063,616
B2 VGG	85.5769%	99.2308%	62.8205%	14.4231%	18,185	19,464,256
B3 ResNet	82.8526%	99.7436%	54.7009%	17.1474%	77,169	25,325,696

FLOPs use one float32 1×1×64×64 evaluation input: 2 per convolution/FC multiply-add plus one per residual addition (28,672 for B3), checked against PyTorch profiler and executed-shape formulas [2]. Exclude BN, activations, pooling, FC bias addition, normalization/transfers, sigmoid/threshold and training work. These consistent scoped estimates are neither total scalar operations nor measured latency.

B1 retains a 4×4 spatial FC head and dropout, but has the most parameters/FLOPs (67,713 head parameters). B2 stacks small kernels with narrower channels/global averaging: fewest parameters/FLOPs, but compressed spatial information and no shortcuts. B3 provides residual gradient routes intended to ease deeper optimization [1], but wider channels/projections cost more than B2 and observed specificity is worse. Shortcut benefits were not isolated.

B2 is only one correct test image ahead of B1; B3 ties B1's sensitivity but makes 16 more false positives. One seed, different batches/selected epochs/update budgets, B1-only dropout and inherited tuning prevent attributing performance differences to architecture alone. The table does not trigger retuning.

[1] He et al. (2016), Deep Residual Learning for Image Recognition.

[2] PyTorch 2.11 profiler documentation (formula-based convolution/matrix FLOP estimates).

B4 · Grad-CAM on five shared random positive examples

NumPy PCG64 seed 42 uniformly samples five label-1 test rows without replacement; IDs were saved before maps, with no visual/prediction filtering. Exact selected checkpoints use final main-path convolutions conv3.conv (B1), conv6.conv (B2), block64.conv2.conv (B3), all under features, before BN/ReLU; B3's shortcut is not separately mapped. Captum targets scalar pneumonia logits (target=None), evaluation mode with gradients enabled: spatially average gradients, weight channels, sum and ReLU [3,4]. Native grids 16×16/16×16/8×8 are bilinearly upsampled to 64×64 and divided by each map's own maximum.

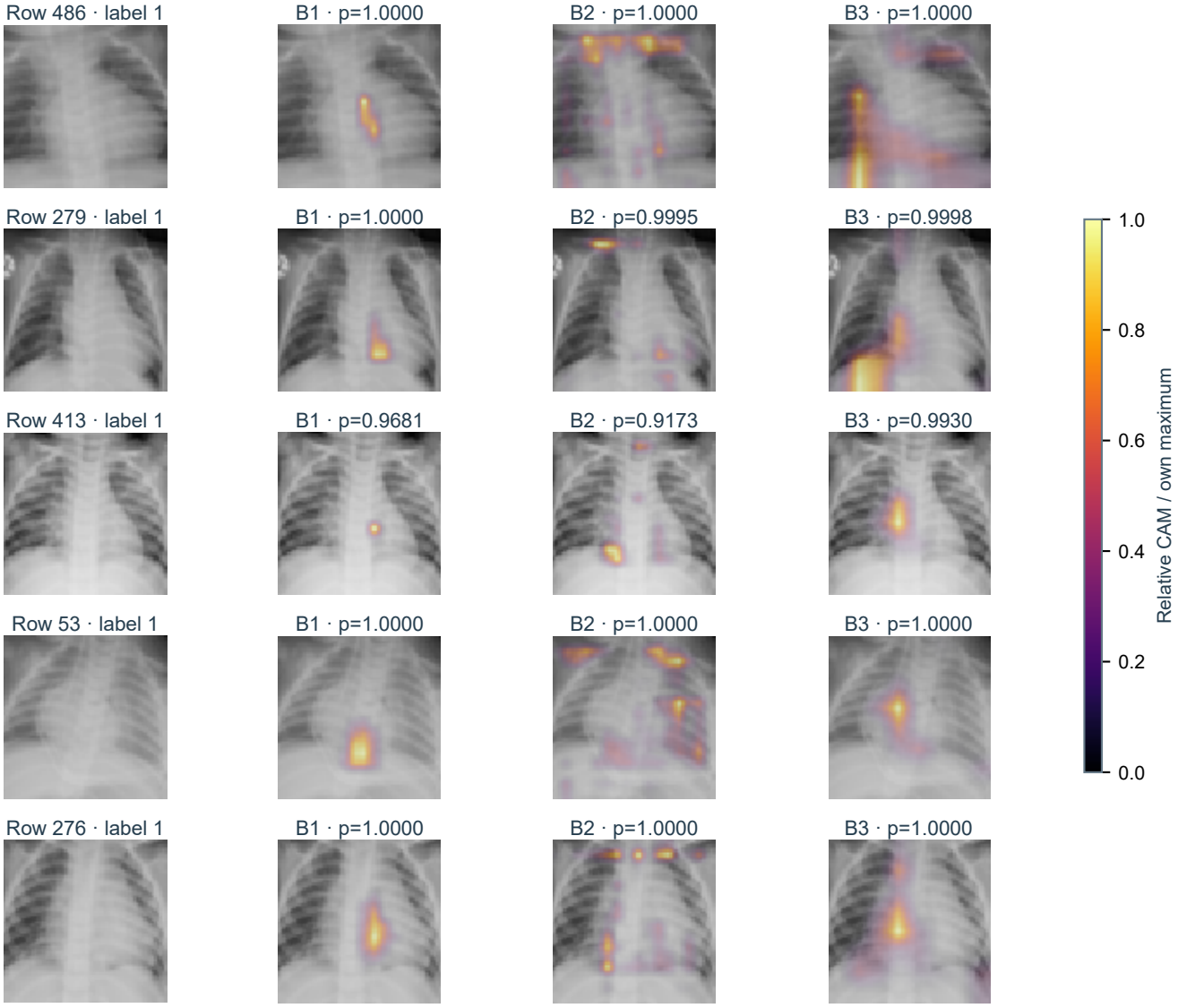


Figure B4. Original pixels at left; shared inferno and opacity=0.55×heat, no flips. p is the saved final-test probability. Each map is normalized separately; colors are relative, not absolute evidence across models.

486: B1 overlaps central brightness; B2 emphasizes the upper edge; B3 the lower image-left boundary. 279: B1 is lower central; B2 upper-edge with weaker lower patches; B3 lower image-left. 413: B1/B3 overlap the bright center, B3 more broadly; B2 peaks lower image-left. 53: B1 is lower central, B2 scattered upper/right, B3 central-left. 276: B1 overlaps the bright central column, B2 favors upper/lower peripheral patches, B3 a broader central-left area. Directions refer to the displayed image.

All five are predicted positive; row 413 is least confident (0.9681/0.9173/0.9930). Coarse maps, normal structures within bright regions and absent lesion annotations prevent validating localization or causal border reliance. Five positive examples do not assess healthy/false-positive behavior. Duplicate selected-input padding matches original CUDA batch sizes; saved logits and independent autograd maps match exactly, with unchanged models and no full-test performance evaluation.

[3] Selvaraju et al. (2017), Grad-CAM: Visual Explanations from Deep Networks.

[4] Captum 0.9.0 LayerGradCam documentation; positive-attribution ReLU is explicit.